

APPENDIX III - SUMMARY PARTICIPANTS INFORMATION SHEET

Resistance Exercise Dosage in Men Receiving ADT for Prostate Cancer

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Participant Information Sheet

SUMMARY

Purpose of the Study: To examine the effects of resistance training or weight training on weight gain, loss of muscle and bone mass, tiredness, immune system and quality of life in men with prostate cancer.

Subject Involvement: We require 60 prostate cancer patients undertaking treatment to suppress male hormones to participate in this 20-week exercise study. Individuals will be required to have non-bone metastases and no musculo-skeletal or cardiovascular disorder that could prevent them from exercising, and not to have participated in any weight training in the previous 12 months. In addition, volunteers will be required to obtain consent from their doctor before participating in the study.

Intervention: Participants will be placed in one of two groups (by chance): immediate exercise or delayed exercise group. Training groups will undertake twice weekly supervised training sessions for 20 weeks. The delayed exercise group will participate in the tests, and receive all information resulting from them, but not exercise during the first 20 weeks of the intervention. A subsequent training program will be provided for the delayed exercise group.

Test Measures: Before starting the training, after 10 and 20 weeks participants will be asked to take part in some tests to measure their strength, muscle characteristics and abilities to take part in activities (i.e. rise from chair, walk 400 meters). Also samples of blood will be obtained to assess relevant characteristics. In addition, participants will be asked to complete some brief questionnaires about their quality of life and abilities to take part in activities of daily living.

PARTICIPATION IS VOLUNTARY AND SUBJECTS ARE FREE TO WITHDRAW FROM THIS STUDY AT ANY TIME.

The study has been approved by Human Research Ethics Committee at Edith Cowan University subjects will be free to withdraw from the study at any time. Contact number, should there be any concerns relating to the project, is (08) 6304 2170 or Email: research.ethics@ecu.edu.au.



APPENDIX IV - PARTICIPANT INFORMATION SHEET

Resistance Exercise Dosage in Men Receiving ADT for Prostate Cancer

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School of Biomedical and Sports Science, Edith Cowan University
Participant Information Sheet

Purpose of the Study

The usual treatments for prostate cancer may have side effects. Some of these side effects include weight gain, loss of muscle and bone mass, tiredness which may affect a person's quality of life. However, exercise has been shown to be a safe and effective method for strength development in older adults. Recently, resistance exercise has also been shown to safely increase muscle strength, quality of life and decrease tiredness in men with prostate cancer. However, the effect of resistance exercise on weight gain, muscle and bone mass and the immune system in this cancer group is unknown. Therefore, the purpose of this present study is to examine the effects of resistance training on weight gain, loss of muscle and bone mass, tiredness, immune system and quality of life in men with prostate cancer. The findings from this study will allow health professionals to plan exercise and the correct amount of exercise for men with prostate cancer to help decrease side effects and improve their quality of life

Subject Involvement

We require 60 prostate cancer patients undertaking treatment to suppress male hormones for at least six weeks to participate in this 20-week exercise study. Individuals will be required to have non-bone metastases and no musculo-skeletal or cardiovascular disorder that could inhibit them from exercising, and not to have participated in any weight training in the previous 12 months. In addition, volunteers will be required to obtain consent from their doctor before participating in the study. In study 1, participants will be randomly allocated (by chance) to one of two groups: immediate exercise or delay exercise group. In study 2, subjects from the immediate exercise group (study 1) will be randomly allocated to two different training protocols: low or traditional training dosage while the delay exercise group will remain in a non-exercise condition. Training groups will undertake twice weekly training for 10 weeks (study 1) with subsequent 10 more weeks (study 2) comprising 20 weeks of intervention. The delay exercise group will participate in the tests, and receive all information resulting from them, but not exercise during the first 20 weeks of the intervention. A subsequent training program based from the outcome measures from the first 20 weeks will be provided for the delay exercise group.

Training protocol

Twice weekly exercise sessions will commence with a 10-minute warm-up comprising of low-level activity such as walking and stationary cycling, as well as stretching.

Eight exercises that target the major upper and lower body muscle groups will be performed using hydraulic resistance machines during the first 10 weeks of training (Study 1). The exercises are: Chest Press/Seated Row, Biceps curl/Triceps extension, Leg Press, Shoulder press/ Lat pull down, Squats, Leg Extension/Leg Curls, Upper row/ Dips, Abdominal/ Back extension.

Ten exercises that target the major upper and lower body muscle groups will be performed using traditional resistance machines in the last 10 weeks of training (study 2). The exercises are: Chest Press, Seated Row, Biceps curl, Triceps extension, Leg Press, Shoulder press, Lat pull down, Leg Extension, Leg Curls, and Abdominal.

Exercise sessions will conclude with a 10-minute cool-down that includes exercises for the abdominal and lower back area and stretching. The exercise sessions will take place in the Strength and Conditioning Laboratory, School of Biomedical and Sports Science, at Edith Cowan University, using hydraulic weight training machines (study

1) and traditional weight training machines (study 2) to ensure participants safety. All sessions will be conducted in small groups of up to 6-8 participants under direct supervision by the principal investigator to ensure proper technique and minimize the risk for injury. The total time to complete the exercise session is less than 60 minutes. All participants will be asked to maintain customary physical activity and dietary patterns over the 20-week period.

The resistance will be set progressively from 12 to 4 repetitions maximum (RM) using repeated lifts before having a rest period. In study 1, the exercise group will perform 2-4 sets of 4-12RM. In study 2, the low dosage training group will perform only 1-2 sets of 4-12RM while the traditional training dosage group will perform 3-5 sets of 4-12RM for each exercise.

To ensure the progressive nature of the training program, both training groups will be encouraged to work past a specific number of repetition-mark. When subjects are able to perform more than the number of repetitions required, during a set the workload will be increased by a 5 - 10% increment for the next set or training session.

All training will take place on Monday and Thursday or Tuesday and Friday.

Test Measures

Prior to the commencement of training, after 10 and 20 weeks a series of tests will be administered. The delay exercise group will only undertake the testing measures before and after the first 20 weeks of intervention and then at 10 and 20 weeks of their exercise intervention.

Body composition and bone density

- Body composition (fat mass and lean mass) and bone density of the hip and spine sites will be measured by dual x-ray absorptiometry (DXA), a routine technique for the measurement of bone density. You will lie on a specially designed table for approximately 10 minutes and a scanning arm will move above your total body and above your hip and spine (separate scans for your whole body, hip and spine). A low-dosage x-ray will pass from underneath the table to the scanning arm. The total radiation dose for all scans undertaken during the study is very low only a little more than normal background radiation and much less than, for example, an

international flight. Note: height and hip/spine density will not be measured at 6 weeks, only before at week 10 and following the 20-week intervention period.

Muscle thickness

- Muscle thickness of the elbow and knee flexors and extensors will be measured by ultrasound images; this is a safe and non-invasive method to assess soft tissues. The scanning probe will be coated with a water-soluble transmission gel to provide acoustic contact to the surface and some ultrasound images will be recorded from the muscles to determine thickness of the muscles. It will take about 5 minutes to obtain all images, the images will be taken when you sit on a chair or stand. All the testing will take place in the Exercise Physiology Research Laboratory in building 19 on the Joondalup Campus.

PSA, Hormones, Immune system, Bone markers and Beta amyloid (A β)

- PSA, Hormones, Immune system markers, Bone markers and A β will be determined from blood samples to assess any changes in these physiological markers. All the testing will take place in the Exercise Physiology Research Laboratory in building 19 on the Joondalup Campus.

Muscle performance

- Maximal muscle strength will be determined for each of the several upper and lower body muscle exercises to be undertaken in the training program using weight-training machines. The maximal strength is the most weight that can be lifted one time using correct technique. This test will also be performed at week 6 and 18 for the immediate exercise group.
- To determine muscle endurance of the upper and lower body, the number of times you can lift a weight that is 70% of your 1RM will be determined.

Functional performance (only undertaken before at week 10 and after the 20-week period)

- Stair climb
You will be asked to ascend a flight of stairs (11 stairs) as fast as possible.
- 6-meter backwards walk
As a test of balance, participants will place one foot well behind the other and will be asked to walk backwards 6 meters.
- Chair rise

You will be seated in a chair and asked to rise and sit 5 consecutive times, without the use of your arms for support, as fast as possible.

- 400-meter corridor walk

You will be asked to walk 20 meters in a corridor, turn and return to the starting position and repeat another 9 times.

- Balance test

You will be asked to stand on a special platform under conditions of eyes open and eyes closed. The platform will tilt slightly back and forth and the surrounding walls will also move. You will be wearing a harness which will support your body weight if you lose balance.

Before muscle performance and physical function tests are performed, demonstrations, practice time, sufficient warm-up and stretching will be undertaken.

Lifestyle questionnaires (only undertaken before and after the 20-week period)

- Quality of life will be assessed using a standardized questionnaire (Functional Assessment of Cancer Therapy-Prostate, FACT-P) as well as Levels of Fatigue (Functional Assessment of Cancer Therapy-anemia, FACT-an) that you can complete at home.
- Health history and Demographic information will be also assessed using standardized questionnaires that you can complete at home

Risks

Resistance training may result in mild discomfort and muscle soreness, however, this will be minimised by all sessions being supervised and commencing with a warm-up and concluding with a cool-down period of mild stretching activities. It is also possible that some muscle soreness may result from baseline performance testing, however, all participants will undertake a warm-up period of stretching before beginning the full exercise program. The risk of discomfort and muscle soreness will also be minimised by a gradual increase in exercise intensity. Lastly, risk of falling may exist in the performance of some tasks, however, participants will be closely supervised and spotted to prevent a fall from occurring. In the event that an emergency occurs, medical assistance will be obtained from the University Health Service according to our established emergency procedures. Lastly, the discomforts

associated with the blood drawing procedures are minimal. There is a risk that sometimes bruising and infection may occur and that the arm might become sore. Risk of bruising or infection from the blood draws will be minimized because all blood draws will be performed by a trained phlebotomist (lab technician) with extensive experience in both research and clinical settings. The total amount of blood drawn during each testing session will not exceed 10 ml. No syringes, lancets, needles or other devices capable of transmitting infection from one person to another shall be reused. All of these items, which are disposable, will be destroyed after each use. As an additional safeguard in preventing contamination new disposable gloves will be required for all blood draws. All contaminated items will be disposed of promptly in sharps containers

All information will be strictly confidential and kept safely locked in a filing cabinet in the primary investigators office. Should publications result from this study, no reference will be made to any individuals.

On completion of the intervention and measurements, a summary of study and individual results will be made available to all participants.

PARTICIPATION IS VOLUNTARY AND SUBJECTS ARE FREE TO WITHDRAW FROM THIS STUDY AT ANY TIME.

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**APPENDIX V - EXAMPLE OF UPPER AND LOWER BODY
EXERCISES USING THE HYDRAULIC RESISTANCE
EXERCISE EQUIPMENT
(Prostate Cancer Study)**

Shoulder press/Lat pull down exercise



Leg extension/ leg curl exercise



Biceps curl/ triceps extension



Chest press/ seated row

